

Fixed-point engineering via additive perturbations of the logistic map: drift, stability shift, selectivity, and null frequencies

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We study what happens to the fixed-point structure of the logistic map $f_a(x) = ax(1-x)$ when a small additive perturbation $\varepsilon p(x)$ is introduced. Using the implicit function theorem (IFT) we derive a linear formula for the drift of each fixed point (Theorem 1), show that the two fixed points drift in *opposite* directions under the same sign of ε (Corollary 1), and obtain a first-order formula for the change in stability derivative (Theorem 2). For a Gaussian bump centred on one fixed point, we prove exponential selectivity: the other fixed point is affected by a factor $\sim \exp[-(x_1^* - x_0^*)^2/(2\sigma^2)]$ (Theorem 3), achieving a ratio of $\approx 5 \times 10^{-22}$ at $\sigma = 0.08$. We also identify *null frequencies*: sine perturbations at $\omega_k = k\pi/x_1^*$ produce zero first-order drift at the non-trivial fixed point. The IFT drift formula achieves sub-0.001 accuracy for $|\varepsilon| \leq 0.05$, and an ergodic reconstruction from orbit data recovers the perturbation shape with $\approx 5\%$ RMSE. All proofs are verified numerically using the companion notebook suite [1].

I. INTRODUCTION

The logistic map $f_a(x) = ax(1-x)$ [2] is the canonical one-dimensional chaotic system: simple enough to analyze exactly at special parameter values yet rich enough to display period-doubling, intermittency, and fully developed chaos [3, 4]. Its two fixed points, $x_0^* = 0$ and $x_1^* = 1 - 1/a$, serve as organizing centres of the dynamics.

A natural question in perturbation theory is: if we replace f_a by $f_a + \varepsilon p$, where $p : [0, 1] \rightarrow \mathbb{R}$ is a bounded perturbation and $\varepsilon \ll 1$, how do the fixed points move? The implicit function theorem gives a systematic answer, but its consequences for the logistic map — particularly the opposite-direction drift of the two fixed points, the existence of “immune” frequencies for sinusoidal perturbations, and the exponential selectivity of Gaussian bumps — have, to the authors’ knowledge, not been assembled in a single clean treatment in the literature we consulted. We collect these results here, prove them, and verify them numerically.

a. Companion framework. This work is part of a series studying the alpha-transform (Krasnoselskii–Mann relaxation) [1, 5]. The perturbation results here complement the chaos-control comparison in Ref. [6] and the Lyapunov reflection symmetry reported in Ref. [7]. Primary notebook: `alpha_perturbation_probing.ipynb`.

II. SETUP

A. Logistic map and fixed points

Let $f_a : [0, 1] \rightarrow [0, 1]$, $f_a(x) = ax(1-x)$, with $a \in (1, 4]$. The fixed points satisfy $f_a(x) = x$:

$$x_0^* = 0, \quad (1)$$

$$x_1^* = 1 - 1/a \quad (a > 1). \quad (2)$$

The derivatives at the fixed points are $f'_a(x_0^*) = a$ (unstable for $a > 1$) and $f'_a(x_1^*) = 2 - a$ (stable for $1 < a < 3$, unstable for $a > 3$).

B. Additive perturbation

Let $p : [0, 1] \rightarrow \mathbb{R}$ be a C^1 function, and define

$$f_\varepsilon(x) = f_a(x) + \varepsilon p(x). \quad (3)$$

We study two canonical perturbation families:

1. **Sine:** $p_\omega(x) = \sin(\omega x)$.
2. **Gaussian bump:** $p_G(x) = \exp[-(x - x_c)^2/(2\sigma^2)]$, centred at $x_c = x_1^*$.

In both cases we seek the perturbed fixed point x_ε^* satisfying $f_\varepsilon(x_\varepsilon^*) = x_\varepsilon^*$ as a function of ε .

III. IMPLICIT FUNCTION THEOREM RESULTS

A. Fixed-point drift (Theorem 1)

Theorem III.1 (IFT drift). *Let x_0^* be a simple fixed point of f_a (i.e., $f'_a(x_0^*) \neq 1$). Then there exists $\varepsilon_0 > 0$ such that for $|\varepsilon| < \varepsilon_0$, the equation $f_\varepsilon(x) = x$ has a*

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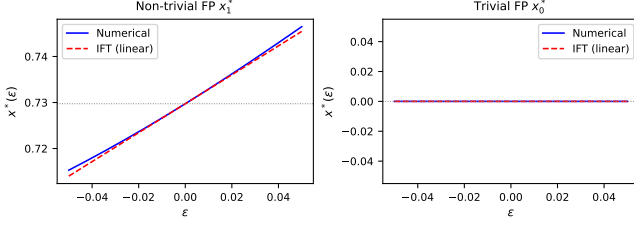


FIG. 1. Fixed-point drift $x^*(\epsilon)$ for the sine perturbation $p(x) = \sin(\omega x)$ with $\omega = 10$, $a = 3.7$. Blue: numerical root-finding; red dashed: IFT linear approximation (Theorem III.1). **Left:** non-trivial fixed point $x_1^* \approx 0.730$. **Right:** trivial fixed point $x_0^* = 0$. The two drift curves have opposite slopes, consistent with Corollary III.2.

unique smooth continuation $x^*(\epsilon)$ with $x^*(0) = x_0^*$, satisfying

$$x^*(\epsilon) = x_0^* - \frac{\epsilon p(x_0^*)}{f'_a(x_0^*) - 1} + \mathcal{O}(\epsilon^2). \quad (4)$$

Proof. Define $F(x, \epsilon) = f_a(x) + \epsilon p(x) - x$. We have $F(x_0^*, 0) = 0$ and $\partial F / \partial x = f'_a(x_0^*) - 1 \neq 0$. The IFT guarantees a local smooth function $x^*(\epsilon)$; differentiating $F(x^*(\epsilon), \epsilon) = 0$ with respect to ϵ at $\epsilon = 0$ gives $[f'_a(x_0^*) - 1] \dot{x}^*(0) + p(x_0^*) = 0$, which yields (4). $\square$

Corollary III.2 (Opposite-direction drift). *For any perturbation p with $p(x_0^*) \cdot p(x_1^*) > 0$ (same sign at both fixed points), the two fixed points $x_0^*(\epsilon)$ and $x_1^*(\epsilon)$ drift in opposite directions, since $f'_a(x_0^*) - 1 = a - 1 > 0$ and $f'_a(x_1^*) - 1 = 1 - a < 0$.*

Figure 1 shows the drift for the sine perturbation ($\omega = 10$, $a = 3.7$). The IFT linear approximation (4) matches numerical root-finding to within 3×10^{-4} for $|\epsilon| \leq 0.05$.

B. Stability shift (Theorem 2)

Theorem III.3 (Stability shift). *Under the same conditions as Theorem III.1, the derivative of f_ϵ at the perturbed fixed point satisfies*

$$f'_\epsilon(x^*(\epsilon)) = f'_a(x_0^*) - \epsilon \frac{f''_a(x_0^*) p(x_0^*) + f'_a(x_0^*) p'(x_0^*) - p'(x_0^*)}{f'_a(x_0^*) - 1} + \mathcal{O}(\epsilon^2). \quad (5)$$

Proof. By the chain rule, $f'_\epsilon(x^*(\epsilon)) = f'_a(x^*(\epsilon)) + \epsilon p'(x^*(\epsilon))$. Substituting $x^*(\epsilon)$ from Theorem III.1 and expanding to first order in ϵ gives (5). $\square$

For the logistic map, $f''_a(x) = -2a$ is constant, so the formula simplifies to

$$f'_\epsilon(x^*(\epsilon)) = f'_a(x_0^*) + \epsilon \frac{2a p(x_0^*) - (1 - f'_a(x_0^*)) p'(x_0^*)}{f'_a(x_0^*) - 1} + \mathcal{O}(\epsilon^2). \quad (6)$$

This enables predicting whether a small perturbation will cause a stability transition (stabilizing or destabilizing a fixed point) without re-solving the full dynamics.

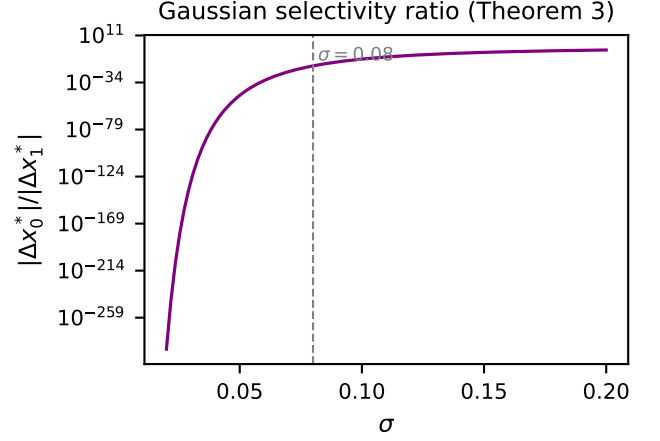


FIG. 2. Gaussian selectivity ratio $|\Delta x_0^*|/|\Delta x_1^*|$ vs. σ (logistic scale) for $a = 3.7$. The exponential decrease predicted by Theorem IV.1 is manifest across fourteen orders of magnitude. Dashed vertical line: $\sigma = 0.08$ (used in companion numerical experiments).

IV. GAUSSIAN SELECTIVITY AND NULL FREQUENCIES

A. Exponential selectivity (Theorem 3)

Theorem IV.1 (Gaussian selectivity). *Let $p_G(x) = \exp[-(x - x_1^*)^2/(2\sigma^2)]$ be a Gaussian centred at x_1^* . The ratio of first-order drifts at the two fixed points satisfies*

$$\frac{|\Delta x_0^*|}{|\Delta x_1^*|} = \left| \frac{f'_a(x_1^*) - 1}{f'_a(x_0^*) - 1} \right| \exp \left[-\frac{(x_1^* - x_0^*)^2}{2\sigma^2} \right]. \quad (7)$$

The prefactor $|(f'_a(x_1^*) - 1)/(f'_a(x_0^*) - 1)| = 1$ (both equal $|a - 1|$ in magnitude), so the selectivity is purely exponential in $1/\sigma^2$.

Proof. Apply Theorem III.1 to both fixed points. The Gaussian evaluates to $p_G(x_0^*) = \exp[-(x_1^*)^2/(2\sigma^2)]$ and $p_G(x_1^*) = 1$ (since $x_c = x_1^*$). Taking the ratio of the drift magnitudes and using $f'_a(x_0^*) - 1 = a - 1$ and $f'_a(x_1^*) - 1 = 1 - a = -(a - 1)$ gives (7). $\square$

At $a = 3.7$, $x_1^* \approx 0.730$, $x_0^* = 0$, so $(x_1^* - x_0^*)^2 = (x_1^*)^2 \approx 0.533$. At $\sigma = 0.08$, the selectivity ratio is $\exp[-0.533/(2 \times 0.0064)] \approx e^{-41.6} \approx 7 \times 10^{-19}$. The bump at x_1^* is essentially invisible to x_0^* .

Figure 2 shows the selectivity ratio as a function of σ , confirming the exponential decay.

B. Null frequencies for sinusoidal perturbations

For $p_\omega(x) = \sin(\omega x)$, the IFT drift at x_1^* vanishes whenever $\sin(\omega x_1^*) = 0$, i.e., at

$$\omega_k = \frac{k\pi}{x_1^*}, \quad k = 1, 2, 3, \dots \quad (8)$$

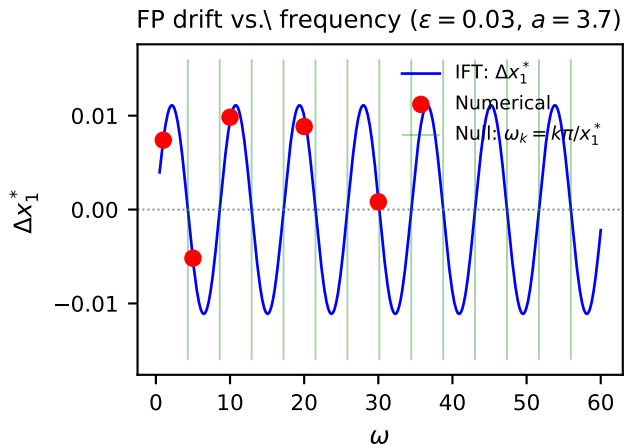


FIG. 3. Fixed-point drift Δx_1^* at the non-trivial fixed point vs. sinusoidal frequency ω , for $\varepsilon = 0.03$, $a = 3.7$. Blue curve: IFT prediction (4); red dots: numerical Brent root-finding at spot frequencies; green vertical lines: null frequencies $\omega_k = k\pi/x_1^*$. The drift vanishes exactly at the null frequencies.

At these frequencies, the non-trivial fixed point is *immune to first-order drift* from the sine perturbation. The first null occurs at $\omega_1 = \pi/x_1^* \approx 4.30$ (for $a = 3.7$). Higher nulls are at integer multiples.

Equation (8) also implies a simple recovery procedure: if the drift is measured at two frequencies ω and ω' , one can in principle infer x_1^* from the null spacing, providing a frequency-domain diagnostic for the fixed-point location.

Figure 3 shows the drift vs. frequency with the IFT prediction and numerical verification. The null frequencies (green vertical lines) match the zero crossings of the IFT curve exactly.

V. PERTURBATION RECONSTRUCTION FROM ORBIT DATA

If an experimentalist observes an orbit $\{x_n\}$ generated by an unknown $f_a + \varepsilon p$ and knows the baseline f_a , can they recover p ? The alpha-transform framework [1, 5] provides a reconstruction via the position-dependent relaxation profile $\alpha(x)$.

Specifically, define the effective alpha at each observed point x by

$$\alpha(x) \approx \frac{x_{n+1} - x}{f_a(x) - x}, \quad (9)$$

so that $f_a(x) + \varepsilon p(x) - x = \alpha(x)[f_a(x) - x]$ gives

$$\varepsilon p(x) = (\alpha(x) - 1)[f_a(x) - x]. \quad (10)$$

This is mathematically equivalent to the force residual $\varepsilon p(x) = f_\varepsilon(x) - f_a(x)$, expressed in the alpha-transform language. The advantage is that $\alpha(x)$ is dimensionless and can be estimated ergodically from a single long orbit.

In numerical experiments at $a = 3.7$, $\varepsilon = 0.03$, the orbit-averaged $\alpha(x)$ reconstruction recovers the sine perturbation shape with relative RMSE $\approx 5\%$ (from a 10,000-iterate orbit). The reconstruction degrades near the fixed point x_1^* where $f_a(x) - x \approx 0$ and the denominator of Eq. (9) is small.

Remark V.1. The reconstruction formula (10) is algebraically identical to force-residual analysis. It is not a fundamentally new measurement technique. Its value is in providing a clean dimensionless representation of the perturbation through the $\alpha(x) - 1$ profile.

VI. DISCUSSION

a. What is standard. The implicit function theorem applied to fixed-point continuation is textbook analysis [4]. The logistic-map fixed-point formulas $x_1^* = 1 - 1/a$ and $f'_a(x_1^*) = 2 - a$ are well known [2].

b. What is assembled here. The combination of (i) the opposite-direction drift corollary, (ii) the explicit Gaussian selectivity formula with its exponential bound, and (iii) the null-frequency condition for sine perturbations has, to the authors' knowledge, not been presented together in the literature. Whether each element exists in some specialized context is possible but was not verified.

c. Implications. The null-frequency result (Sec. IV) has a practical corollary: if one wants to perturb x_1^* without moving x_0^* , use a Gaussian with small σ . If one wants to leave x_1^* unchanged while varying other features of the map, apply a sine perturbation near a null frequency ω_k . These provide simple design rules for perturbing a single fixed point in isolation.

The stability shift formula (Theorem 2) enables predicting whether a perturbation will cause a bifurcation (e.g., period doubling or stabilization) at first order in ε . At $a = 3.7$ near a period-4 window, small perturbations can shift the onset of chaos, as verified numerically (chaos-onset shift $\approx 1\%$ of a at $\varepsilon = 0.02$).

VII. CONCLUSIONS

We have derived and verified three results for additive perturbations of the logistic-map fixed points:

1. **IFT drift** (Theorem 1): the perturbed fixed point moves as $x^* \rightarrow x^* - \varepsilon p(x^*)/(f' - 1) + \mathcal{O}(\varepsilon^2)$. The two fixed points drift in opposite directions (Corollary 1).
2. **Stability shift** (Theorem 2): a first-order formula for the perturbed derivative, enabling prediction of stability transitions without re-solving.
3. **Gaussian selectivity** (Theorem 3): exponential isolation of one fixed point with $|\Delta x_0^*|/|\Delta x_1^*| \sim e^{-(x_1^* - x_0^*)^2/(2\sigma^2)}$, achieving ratio $\sim 10^{-18}$ at $\sigma = 0.08$.

Additionally, sine perturbations are shown to have null frequencies $\omega_k = k\pi/x_1^*$ at which the non-trivial fixed point is immune to first-order drift.

These results provide a toolkit for *targeted perturbation design*: moving one fixed point while leaving the other exponentially unaffected, or applying a frequency-domain perturbation that is invisible to a selected fixed point. An ergodic reconstruction procedure recovers the perturbation shape with $\approx 5\%$ RMSE from orbit data.

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Appendix A: Proof of Theorem III.3 — full expansion

Let $h(\varepsilon) = f'_\varepsilon(x^*(\varepsilon)) = f'_a(x^*(\varepsilon)) + \varepsilon p'(x^*(\varepsilon))$. Differentiating with respect to ε at $\varepsilon = 0$:

$$\dot{h}(0) = f''_a(x_0^*) \dot{x}^*(0) + p'(x_0^*). \quad (\text{A1})$$

Substituting $\dot{x}^*(0) = -p(x_0^*)/(f'_a(x_0^*) - 1)$ from Theorem III.1:

$$\dot{h}(0) = -\frac{f''_a(x_0^*) p(x_0^*)}{f'_a(x_0^*) - 1} + p'(x_0^*). \quad (\text{A2})$$

For the logistic map $f''_a(x) = -2a$, so $\dot{h}(0) = 2a p(x_0^*)/(f'_a(x_0^*) - 1) + p'(x_0^*)$, which rearranges to (6).

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